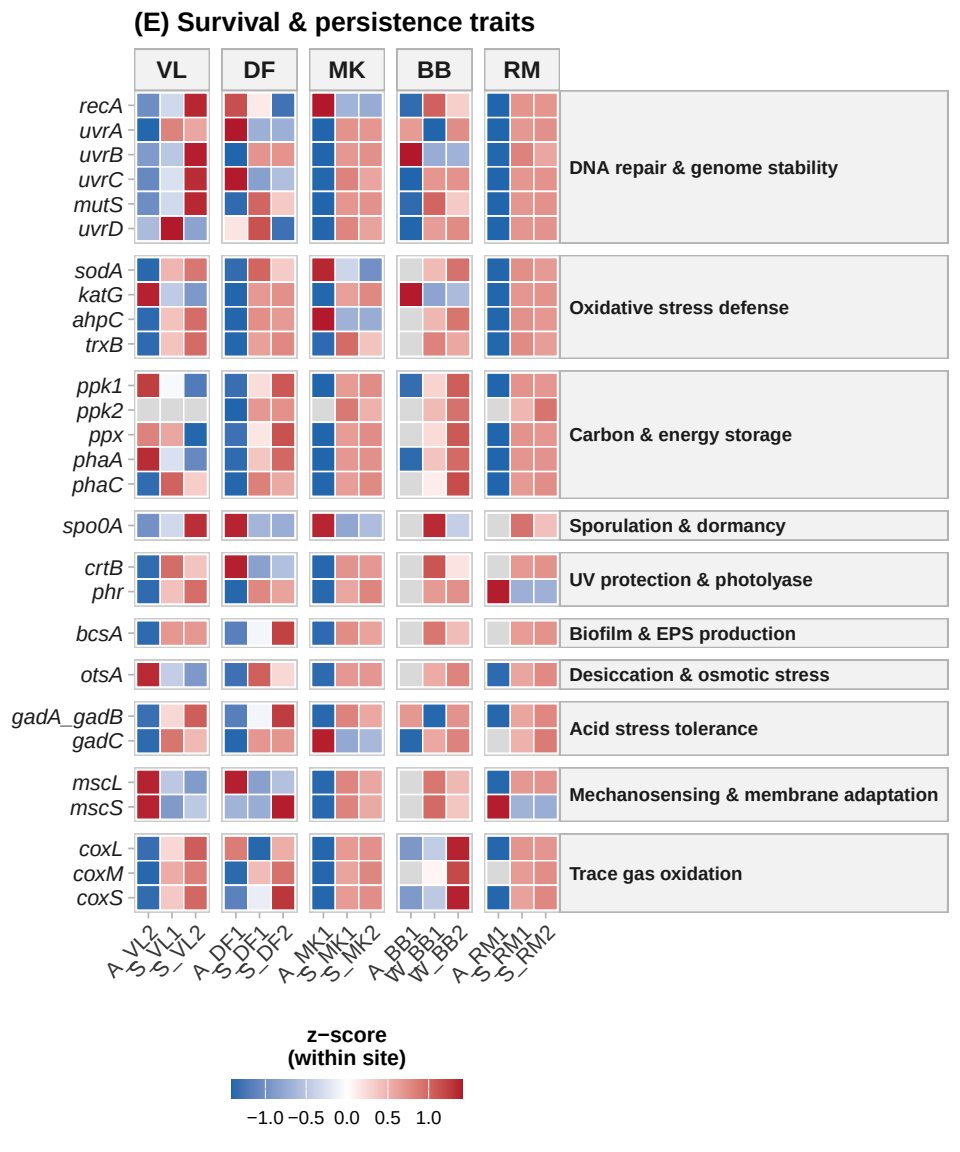
(E) Survival & persistence traits
VL
DF
MK
BB
RM
recA
uvrA
uvrB
uvrC
mutS
uvrD
DNA repair & genome stability
sodA
katG
ahpC
trxB
Oxidative stress defense
ppk1
ppk2
ppx
phaA
phaC
Carbon & energy storage
spo0A
Sporulation & dormancy
crtB
phr
UV protection & photolyase
bcsA
Biofilm & EPS production
otsA
Desiccation & osmotic stress
gadA_gadB
gadC
Acid stress tolerance
mscL
mscS
Mechanosensing & membrane adaptation
coxL
coxM
coxS
Trace gas oxidation
A_VL2
S_VL1
S_VL2
A_DF1
S_DF1
S_DF2
A_MK1
S_MK1
S_MK2
A_BB1
W_BB1
W_BB2
A_RM1
S_RM1
S_RM2
z−score
(within site)
−1.0 −0.5 0.0 0.5 1.0